

Entity' button, and the tool proceeds to warn you and perform initialisation (Figures 4 and 5). Once initialised, the partition is moved into the 'Unallocated Volumes' (Figure 6) branch. Here, click the 'Add to Existing Volume Group' button, which prompts you to select the volume group (VG) to which you will add this partition. This adds unused space to your 'ubuntu' VG (Figure 7). Then edit an LV, and add this unused space into it (Figure 8). Select the LV adjacent to the unused space, and hit the *Edit* button. Note that you can now extend the volume by



Figure 8: Editing a Logical Volume

dragging the slider. Drag as required, and hit 'OK'! This will add the unused space to your LV.

This is how LVM configuration is done using the GUI. The command-line provides you with a comprehensive set of commands to do this, but it's risky for a new user, since a wrong command can cause you to lose data.

LVM is very useful, since you can add, extend and remove volumes irrespective of the physical disks in the system; you can combine disks to appear as single drives. LVM eases upgrading and extending disk capacities. Fundamentally, working with LVM is almost the same as working with regular partitions; the only difference is that in the case of LVM, we deal with a 'logical' space and its relation to 'physical' space—also known as storage virtualisation. I hope this article has whetted your appetite to explore storage technologies in greater depth. Have fun! 

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Continued from page 66....

Virtualisation layer	Security implementation
Physical host	- Physical security - Firewall - Intrusion Detection System
Host OS	- OS hardening - Access control
Virtual HAL	- Install from genuine source - Avoid stray instances - Apply security patches
Guest OS	- OS hardening - Access control - Patching
Guest application	Application-level security installations and controls

Implementing security for a virtualisation infrastructure is a methodical and multi-step process. We need to begin with the physical infrastructure, followed by the installation and the hardening of OS on it. If Xen server is being installed from scratch, getting an infection-free copy of the distribution is critical. The next step is to configure user-level security, enabling access only to administrators, to create, delete or modify the guest OSs. It should be ensured that no stray guest instances are left lingering on the infrastructure, and should be deleted if found. While configuring guest OSs, it is important to remember that

each instance may run a different make or version. Hence, a unified patch-management solution that can cater to heterogeneous systems needs to be deployed. The same is true for the anti-virus software.

Besides patching, each OS should be hardened and configured for correct security accesses. As mentioned earlier, the communication between guest instances is not visible on the physical network interface; hence, an intrusion detection system should be deployed. The correct way to do this is to dedicate one guest instance on each virtual infrastructure for the purpose of packet sniffing and intrusion detection. Physical security for virtual servers is very crucial, because stealing a physical machine means stealing more than one VM, thus resulting in serious data theft or infrastructure breakdown. For VMs hosted in the demilitarised zone, a proper perimeter defence system comprising firewalls and UTM devices should be deployed too. 

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